# Interpretable and Robust Multimodal Remote Sensing Image Matching via Phase Cue Feature Learning

Peifu Lai, Yaping Yan, Na Zhang, and Songlin Du, *Senior Member, IEEE*

***Abstract*—Since the complementary information offered by different modalities is conducive to generating more comprehensive representation for relevant applications, multimodal remote sensing image matching has garnered growing research attention. However, due to the severe nonlinear radiation differences (NRDs) between different kinds of images, it is still a challenging task. Despite the viable solutions proposed by advanced spatial-domain methods and frequency-domain methods, the integration of these two domains information still remains to be explored. In this article, we propose a multimodal remote sensing image matching method via phase cue feature learning that couples phase perception detection and description (P2D2). By combining the advantages of spatial and frequency domain analyses, P2D2 establishes a novel framework for interpretable and robust cross-modal feature learning. First, P2D2 uses phase image (PI) extractor to produce the PIs of respective modality images, which are utilized to explore the shared frequency-domain information between different modalities images. After leveraging an unshared lightweight tri-branch encoder to process input images, the phase perception detector (PPDet) is designed to extract more repeatable and interpretable keypoints. Then, P2D2 designs the diverse structural information augmented descriptor (DSIADes) to integrate the discriminative features from images of respective modality and the various structural features provided by PIs. In addition, the mutual weighted losses are designed to further improve the repeatability of keypoints and the robustness of modality-invariant feature representations. The experimental results demonstrate the effectiveness of P2D2 in both feature detection and description. Compared with seven state-of-the-art methods, including locally normalized image feature transform (LNIFT), radiation-variation insensitive feature transform (RIFT), weighted structure saliency feature (WSSF), ReDFeat, and attention-based feature (AttFeat), P2D2 is the only one that achieves a matching success rate (MSR) of 100% across all four representative datasets and consistently demonstrates superior cross-dataset generalization.**

Received 22 June 2025; revised 14 November 2025 and 11 January 2026; accepted 10 February 2026. Date of publication 17 February 2026; date of current version 5 March 2026. This work was supported in part by National Natural Science Foundation of China under Grant 62201142, in part by the Shenzhen Science and Technology Program under Grant JCYJ20240813161801002, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2025A1515011943, and in part by the Foundation of National Key Laboratory of Complex System Control and Intelligent Agent Cooperation under Grant 2024-002. *(Corresponding author: Songlin Du.)*

Peifu Lai and Songlin Du are with the School of Automation, Southeast University, Nanjing 210096, China, and also with the Shenzhen Research Institute of Southeast University, Shenzhen 518063, China (e-mail: laipeifu@seu.edu.cn; sdu@seu.edu.cn).

Yaping Yan is with the School of Computer Science and Engineering, Southeast University, Nanjing 211189, China (e-mail: yan@seu.edu.cn).

Na Zhang is with Beijing Electro-Mechanical Engineering Institute, Beijing 100074, China (e-mail: 1428469148@qq.com).

Digital Object Identifier 10.1109/TGRS.2026.3665603

***Index Terms*—Multimodal image analysis, phase cue feature learning, remote sensing image matching.**

## I. Introduction

MULTIMODAL remote sensing image matching generally refers to the process of establishing pixelwise correspondences from the same scene images acquired by different sensors [1], such as optical and synthetic aperture radar (SAR) images. It is a fundamental but crucial task in the field of image fusion and enhancement [2], [3], change detection [4], [5], visual localization, and navigation [6], [7].

In the past few decades, many efforts have been made to solve the problem of image matching, and various well-known methods have been proposed. For example, many handcrafted methods, such as scale-invariant feature transform (SIFT) [8], speeded-up robust features (SURFs) [9], and oriented FAST and rotated BRIEF (ORB) [10] are widely used in the fields of robot vision due to their cost-effectiveness. With the development of deep learning, various learning-based methods have also sprung up. SuperPoint [11] first presents a self-supervised framework for detection and description, which is suitable for most geometric computer vision tasks. To increase the reliability of the detected features, R2D2 [12] proposes a fully-convolutional network (FCN) for jointly learning repeatable and reliable matches by learning an extra reliability map. Besides, various advanced methods are proposed for computational efficiency, such as ALIKE [13], SiLK [14], and XFeat [15]. However, most of these methods focus on the same-source images matching and are not suitable for multimodal image matching.

Compared to unimodal images, multimodal remote sensing images contain complementary information, which is beneficial to yield a more comprehensive representation for related applications. In particular, an optical camera is capable of capturing both spectral and spatial information, while SAR offers robust all-day and all-weather imaging with cloud penetration capabilities. By matching optical and SAR images, it can easily gather the advantages of different modalities and realize information complementation to mitigate information loss [16]. Nevertheless, due to their different imaging mechanisms, there are significant appearance differences between multimodal remote sensing images. Severe nonlinear radiation differences (NRDs), complex geometric differences, and noise interference make the matching task much more challenging.

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

To navigate these challenges, scholars have made great efforts to study multimodal remote sensing image matching and have proposed many methods. For example, as the cross-modal counterparts of SIFT, optical-SAR SIFT (OS-SIFT) [17] comprehensively considers the inherent properties of optical and SAR images, refining SIFT for optical-to-SAR image registration. Similarly, Several methods are also proposed for heterogeneous feature matching, such as partial intensity invariant feature descriptor (PIIFD) [18], position-scale-orientation SIFT (PSO-SIFT) [19], and nonlinear diffusion SIFT (ND-SIFT) [20]. However, these methods can only achieve remarkable performance on specific image types, which inherently restricts their applicability beyond the defined feature space.

To solve this problem, several frequency-domain methods were proposed. Ye and Shen [21] focused on the structural properties of images and proposed the histogram of oriented phase congruency (HOPC). Li et al. [22] proposed the radiation-variation insensitive feature transform (RIFT) to adaptively eliminate severe NRD between different types of multimodal image pairs, such as optical-infrared (IR), optical-SAR, and optical-depth. Nonetheless, these methods based on frequency-domain information suffer from high computational overhead and memory consumption. Hence, Li et al. [23] proposed the locally normalized image feature transform (LNIFT) to mitigate the modal variance and achieve nearly real-time processing performance. To address the problem of sensitivity to complex geometric variances, co-occurrence filter space matching (CoFSM) [24] and multiscale histogram of local main orientation (MS-HLMO) [25] were proposed to achieve rotation and scale invariance of multimodal image matching. Furthermore, Li et al. [26] proposed a new scale and rotation invariant feature transform (SRIF) method for multimodal image matching to deal with these two types of geometric changes simultaneously.

In recent years, various deep learning techniques have been effectively introduced into the field of multimodal remote sensing image matching. Benefiting from the development of deep learning technology, learning-based methods achieve better matching performance over traditional handcrafted methods. For instance, siamese convolutional neural network was introduced to handle the task of optical-SAR matching [27], [28]. CMM-Net [29] and ReDFeat [30] both use a pseudo-Siamese network to learn modality-invariant feature representation, which consists of a shallow two-branch layer without sharing parameters to mitigate the variance of different modalities and a weight-sharing layer embedded into the deep layers of the network to extract the high-level semantic information. Besides, various learning-based methods [31], [32], [33], [34], [35] focused on the nonlinear radiometric differences and geometric distortion between multimodal images also achieve promising results. Nevertheless, it is still challenging to learn the deep features with both the modal-invariant information and the modal-specific information.

Existing works have shown that frequency domain analysis is pivotal to learning the shared information in cross-modal images [36], [37], [38]. In frequency domain analysis, the Fourier transform is commonly utilized to decompose the image into its constituent phase and amplitude components. Notably, the structural information corresponding to phase components is proven to be much more robust to noise and brightness changes [39], [40]. To enhance the performance of multimodal image matching, recent studies have begun to further investigate the useful frequency components present across different modal images. Zhang et al. [41] proposed a histogram of the orientation of weighted phase (HOWP) to resolve the problems of directional inversion or phase extreme value mutations in phase feature calculation. Wan et al. [42] proposed a weighted structure saliency feature (WSSF) using structure saliency information by combining the second-order Gaussian steerable filtering, edge confidence map, and phase features. Quan et al. [43] proposed an adaptive frequency feature filtering network (F3Net), which performs frequency filtering on multilevel deep features to explicitly explore the useful frequency information.

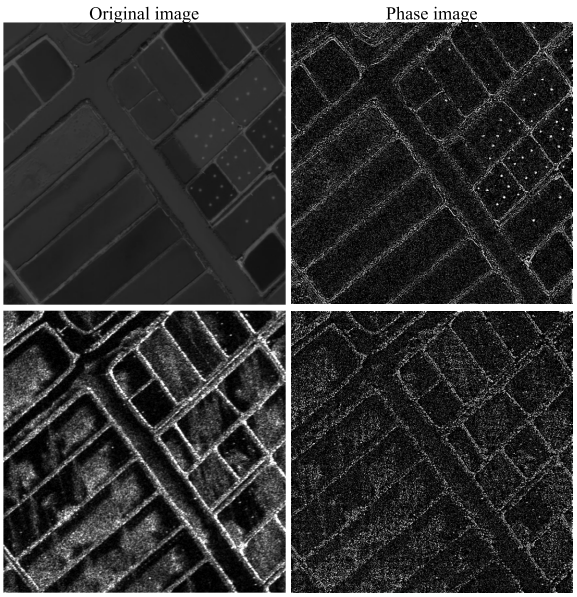



Fig. 1. Comparison of original images with their PIs.

According to [44] and [45], high-level image content and semantic information mainly reside in the high-frequency phase components from the Fourier transformed frequency domain, and the distortions applied to the low-frequency phase and amplitude components hardly affect the modality-invariant features. As shown in Fig. 1, the NRDs between phase images (PIs) are smaller, and the structural properties of PIs are more significant than original cross-modal images, which is conducive to both keypoint detection and description. Inspired by this, this article proposes a multimodal remote sensing image matching method via phase cue feature learning and establishes a joint detection and description framework, called phase perception detection and description (P2D2). The proposed method enables the detector to locate the salient regions in cross-modal images, such as corners, lines, or blobs. In addition, P2D2 can facilitate the integration of diverse surrounding structural information into local descriptors.

To accomplish the objective, P2D2 first applies the PI extractor to produce the PIs of respective modality images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

Then, P2D2 uses an unshared lightweight tri-branch encoder to mitigate the gap of multimodal images. After that, we improve our detector in terms of both network structure and network optimization. From the perspective of network structure, the phase perception detector (PPDet) consists of four-phase perception blocks and a differentiable nonmaximum-suppression (DNMS) unit. Instead of the standard direct skip-connections, P2D2 employs adaptive attention gates within the expanding path to generate multiscale phase masks that allow PPDet to extract homonymous points from cross-modal images. From the perspective of network optimization, we propose a keypoint detection loss where a high-frequency prior generated by PIs is introduced to improve the repeatability and interpretability of keypoints. Moreover, P2D2 designs the diverse structural information augmented descriptor (DSIADes) to integrate the modal-specific features of respective modality images and their frequency-domain features of PIs from different receptive fields, which allows the descriptors to fully embrace the multilevel structural awareness and further improve the matching accuracy.

The contributions of this article are summarized as follows.

1) A multimodal remote sensing image matching method named P2D2 is proposed, which focuses on joint feature detection and description based on the integration of spatial-domain and frequency-domain information.
2) This article introduces a PPDet based on a multilevel attention gate mechanism, integrating a detection loss function grounded in high-frequency prior knowledge from PI to enhance keypoints repeatability and interpretability.
3) A DSIADes is proposed to enable local descriptors to make full use of the multilevel surrounding structural information via different depthwise convolutions. However, the spatial-frequency description loss is utilized to further improve the discriminability of descriptors.

## II. Related Work

Multimodal remote sensing image matching can usually be classified into area-based and feature-based methods, which can be briefly reviewed as follows.

### A. Area-Based Methods

Area-based matching typically denotes the process of identifying optimal template positions within reference images through the computation of specific similarity metrics using a sliding window, which is also known as template matching.

The core task of handcrafted area-based matching is to define the similarity metrics. Early template matching methods use the metrics in the spatial domain, such as the sum of square differences (SSD), normalized cross correlation (NCC), and mutual information (MI). To enhance the robustness to NRDs, numerous phase correlation methods have been proposed, which calculate the similarity in the frequency domain based on the Fourier transform. For instance, Ye et al. [21] proposed HOPC and the channel features of oriented gradients (CFOGs) [46], which use the fast Fourier transform (FFT) for fast similarity measure in the frequency domain. Besides, the phase congruency structural descriptor (PCSD) [47] and angle-weighted oriented gradients (AWOGs) [48] are also representative.

Given that deep learning had shown great potential in feature representation, several methods adopted deep learning techniques for multimodal template matching. In particular, Hughes et al. [49] proposed a generative hard-negative mining framework for optical-SAR image template matching. Ye et al. [50] proposed a multiscale framework with unsupervised learning (MU-Net) to directly learn the transformation parameters of remote sensing image pairs. However, these learning-based template matching methods are still restricted by their dependence on training datasets and high computation.

To integrate the strengths from both handcrafted methods and learning-based methods, Ye et al. [51] propose a hybrid matching method using attention-enhanced structural features (AESFs), which combines handcrafted structural feature extraction with attention enhancement in deep learning. Nevertheless, area-based multimodal image matching methods are still very sensitive to local extrema and geometric transformations.

### B. Feature-Based Methods

Feature-based matching consists of feature detection, feature description, and correspondence estimation. First, common salient keypoints with high repeatability are extracted by feature detectors. Then, these keypoints are mapped into the discriminative vectors via feature descriptors. After that, a one-to-one correspondence between two feature sets is established.

To solve the problem of significant NRDs between cross-modal images, numerous handcrafted feature-based methods have been proposed. For example, PIIFD [18] was proposed for retinal image matching. PSO-SIFT [19] and OS-SIFT [17] are developed for heterogeneous feature matching. However, these methods are only suitable for certain image types and suffer from a lack of generalization. Therefore, Li et al. [22] proposed RIFT, which achieves remarkable performance on different types of images. Cui et al. [52] extended the RIFT by adding a scale space stage to achieve scale invariance. Furthermore, LNIFT [23] was proposed to cope with the high computation and memory consumption of the above methods and perform in nearly real-time. Besides, Li et al. [26] proposed SRIF to handle the complex geometric differences, and Wan et al. [42] proposed WSSF using structural information to improve the matching performance.

Recently, learning-based feature matching has achieved great progress in multimodal remote sensing image matching. Hughes et al. [53] first designed a three-stage framework for optical-SAR matching, which fully applies a deep neural network in every stage. Cui et al. [31] proposed a matching method for SAR to optical images based on spatial pyramid aggregated pooling and attention mechanism. CMM-Net [29] and ReDFeat [30] using a pseudo-Siamese network have been proposed for joint detection and description, which both achieve promising performance on multispectral images. Li et al. [35] proposed the attention-based feature (AttFeat) to extract features in sparse and repetitive texture regions of multimodal remote sensing scenes, which utilizes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

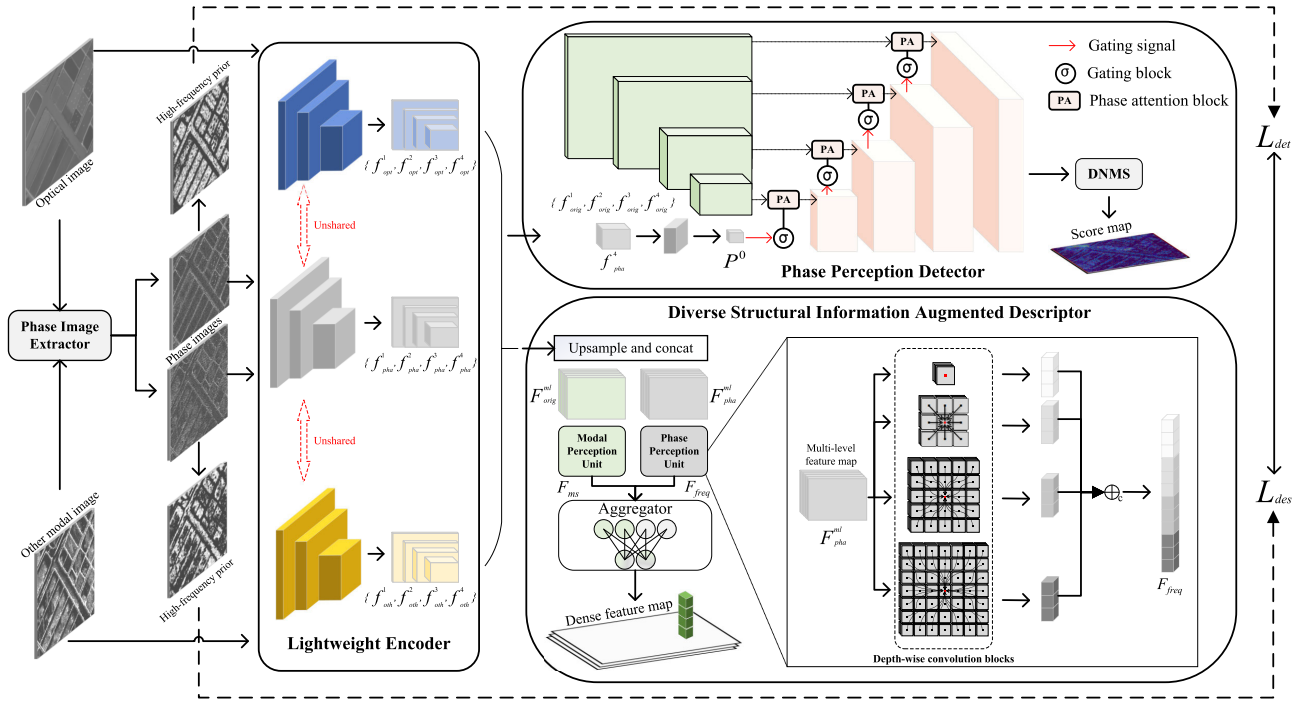



Fig. 2. Proposed P2D2 framework consists of the PI extractor, unshared lightweight tri-branch encoder, PPDet, and DSIADes.

a parallelized structure to combine coordinate, spatial, and channel features. Furthermore, Zhang et al. [34] combined the advantages of deep convolutional neural networks and the transformer attention mechanism and proposed modality-independent consistency matching (MICM) to improve the accuracy of multimodal remote sensing image matching. Besides, Li et al. [54] developed a robust matching method, which utilizes feature enhancement with multilevel attention alternating combinations against the noise interference of multisource remote sensing images.

This article attempts to fully utilize distinctive information in both spatial and frequency dimensions in cross-modal images. We propose a joint detection and description framework named P2D2 via phase cue learning that consists of PPDet, DSIADes, and the mutual weighted losses for multimodal remote sensing image matching.

## III. Methodology

### A. Overview

In this article, a joint detection and description framework named P2D2 is proposed for end-to-end multimodal remote sensing image matching. As shown in Fig. 2, P2D2 produces the PIs of respective modality images by the PI extractor. Then, three types of images will pass through an unshared lightweight tri-branch encoder to reduce significant NRDs. Furthermore, we use the PPDet to extract keypoints in the salient regions and the DSIADes to generate robust features for matching. The details of the proposed method are described as follows.

### B. Network Structure

*1) PI Extractor:* To explore the frequency-domain information of multimodal remote sensing images with a computationally efficient approach, P2D2 first utilizes the PI extractor to generate PIs for respective modality images in the preprocessing stage. Given an input image $I \in \mathbb{R}^{H\times W}$, we adopt the Fourier transform $\mathcal{F}$ to get the frequency representation

$$R_{freq}(u,v) = \mathcal{F}(I)(u,v) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} I(h,w)\, e^{-j2\pi\left(\frac{h}{H}u+\frac{w}{W}v\right)}. \tag{1}$$

According to [21], the structural information of the images is mainly provided by the phase. We can get the corresponding phase component $R_{pha}$ of the input image $I$

$$R_{pha}(u,v) = \arctan\left(\frac{\mathrm{Im}\left(R_{freq}(u,v)\right)}{\mathrm{Re}\left(R_{freq}(u,v)\right)}\right) \tag{2}$$

where the operators Im(•) and Re(•) symbolize the imaginary and real parts of the input.

Then, we apply the inverse Fourier transform to produce the $PI$ of the input image $I$

$$PI = \mathcal{F}^{-1}\left(A_{const}e^{-jR_{pha}(u,v)}\right) \tag{3}$$

where the constant amplitude $A_{const}$ is introduced to control the numerical scale of the inverse Fourier reconstruction.

*2) Unshared Lightweight Tri-Branch Encoder:* To adaptively learn the modal-invariant representation from multimodal remote sensing images and the robust frequency representation from PIs, three kinds of images (optical images, other modal images, and their corresponding PIs) are passed through a lightweight tri-branch encoder, which consists of three residual convolution blocks for each branch. Each residual block has two convolutional layers composed of $3 \times 3$ convolutional operators followed by batch normalization and ReLU activation, and a shortcut path composed of $1 \times 1$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

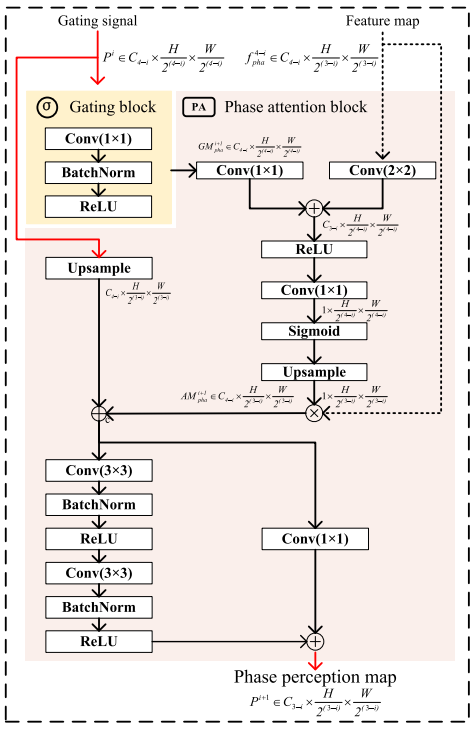



Fig. 3. Details of the gating block and phase attention block.

convolutional operator followed by batch normalization. The max pooling with a kernel size of 2 is adopted between each residual block to decrease the resolution of feature maps and thus reduce the computational cost. Despite adopting an unshared tri-branch design, the proposed lightweight encoder requires less than 5% of the parameters per branch compared to a typical pseudo-Siamese architecture such as CMM-Net [29] while maintaining a low overall parameter complexity.

In particular, for an input image $I \in \mathbb{R}^{H\times W\times 3}$, the first feature map $f^1 \in \mathbb{R}^{H\times W\times 3}$ is generated by

$$f^1 = InsNorm\,(I) \tag{4}$$

where *InsNorm* represents the instance normalization for retaining the original content structure.

Then, $f^2 \in \mathbb{R}^{H/2\times H/2\times 32}$, $f^3 \in \mathbb{R}^{H/4\times H/4\times 64}$, and $f^4 \in \mathbb{R}^{H/8\times H/8\times 128}$ feature maps are produced by three sequential residual block. These feature maps derived from the encoder will be used for subsequent detection and description.

*3) Phase Perception Detector:* To extract high-frequency texture features at multiple levels, P2D2 designs the PPDet based on an attention gate mechanism, which consists of four-phase perception blocks and a DNMS unit.

To locate the salient regions in cross-modal images, PPDet first uses a residual block to further downsample the final feature map of PI $f^4_{pha}$ and generate the phase perception map $P^0 \in \mathbb{R}^{H/16\times H/16\times 128}$. Then, PPDet employs four-phase perception blocks with attention gates to restore the feature map to the original image size and make the feature map much sharper and more focused in salient regions. As shown in Fig. 3, $P^0$ is passed through a gating block that consists of $1 \times 1$ convolution followed by batch normalization and ReLU activation to generate the phase gated mask $GM^1_{pha} \in \mathbb{R}^{H/16\times H/16\times 128}$. After that, the final feature map of the original image $f^4_{orig}$ and the phasegated mask $GM^1_{pha}$ are used to produce the phase attention map $AM^1_{pha} \in \mathbb{R}^{H/8\times H/8\times 128}$ by the phase attention block. Then, we use bilinear interpolation to upsample $P^0$ and concatenate it with the phase attention map $AM^1_{pha}$ along the channel dimension. Finally, the concatenated feature map will pass through a residual block to produce the phase perception map $P^1 \in \mathbb{R}^{H/8\times H/8\times 64}$. Similarly, the phase perception maps $P^2$, $P^3$, and $P^4$ can be obtained by the following phase perception blocks.

To improve the distribution of keypoints and avoid the vanishing of gradient caused by the traditional nonmaximum-suppression (NMS), we incorporate a DNMS unit into our detector. The differentiable NMS unit comprises two collaborative differentiable branches. The first branch consists of $1 \times 1$ convolution, softplus activation, and normalization to highlight potential significant features, while the second branch produces a local suppression weight map via two dilated convolutions, local normalization, and softmax to emphasize dominant features. However, the phase perception map $P^4$ will pass through the DNMS to generate the final score map of detection $S \in \mathbb{R}^{H\times W\times 1}$. Furthermore, to enhance the robustness of scale changes, we downscale the two input images into different sizes, extract keypoints for all scaled image and mine the local maximum of their score maps during the inference phase. Finally, *Top-k* points with the highest detection scores are selected as the final keypoints.

*4) Diverse Structural Information Augmented Descriptor:* To aggregate diverse surrounding structural information from different levels and different receptive fields, P2D2 designs the DSIADes that consists of a modal perception unit (MPU) and a phase perception unit (PPU). In contrast to most previous works that only deploy the top-layer feature maps to construct descriptors, we utilize all feature maps generated by the encoder to mine the textural information from different levels.

In particular, we first use bilinear interpolation to upsample the feature maps $f^2$, $f^3$, and $f^4$ to the size of the original image and then concatenate them to obtain the multilevel feature map $F^{ml} \in \mathbb{R}^{H\times W\times 227}$. For the MPU, the concatenated feature map of the original modal image $F^{ml}_{orig}$ simply passes through three convolutional layers with batch normalization to generate the modal-specific feature map $F_{ms} \in \mathbb{R}^{H\times W\times 128}$ of respective modality images. For the PPU, we construct multiple description spaces to integrate diverse surrounding structural information from the multilevel features of PIs. We decouple the frequency-domain feature map into four 32-dimensional sub-feature maps and enable them to consolidate the surrounding structural information of phase feature maps from different receptive fields. Given the concatenated feature map of PIs $F^{ml}_{pha}$, we use four depthwise convolution blocks with kernel size of 1, 3, 5, and 7, respectively to generate four sub-featuremaps $F^1_{freq}$, $F^2_{freq}$, $F^3_{freq}$, and $F^4_{freq}$ from different receptive fields. The frequency-domain feature map is computed as

$$F_{freq} = F^1_{freq} \oplus_c F^2_{freq} \oplus_c F^3_{freq} \oplus_c F^4_{freq} \tag{5}$$

where $F^i_{freq} \in \mathbb{R}^{H\times W\times 32}$ $(i = 1, 2, 3, 4)$ and the symbol $\oplus_c$ represents the concatenation along the channel dimension.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

Finally, the final dense feature map with modal-specific information and diverse surrounding structural information is obtained by

$$D = \mathcal{F}_{Agg}\left(F_{ms}, F_{freq}\right) \tag{6}$$

where $D \in \mathbb{R}^{H\times W\times 128}$ and $\mathcal{F}_{Agg}$ represent the aggregator, which concatenates feature maps along the channel dimension and subsequently applies the $1 \times 1$ convolutional transformation for cross-channel information interaction and integration.

### C. Network Optimization

To mitigate the discrepancies between multimodal remote sensing images and learn modality-invariant feature representation, we train our P2D2 using a detection loss with a high-frequency prior and a spatial-frequency description loss. The mutual weighting strategy is utilized to couple detection and description for further improvement of feature learning. According to Section III, we can, respectively, sample K pairs of corresponding descriptors from the optical image $I^{opt}$ and the other modal remote sensing image $I^{oth}$ employing the detection score map $S$ and the dense feature map $D$ to construct the detection score set $\{s_i^{opt}, s_i^{oth}\}$ and the corresponding descriptor set $\{d_i^{opt}, d_i^{oth}\}$, where $i = 1, 2, \ldots, K$.

*1) Detection Loss With High-Frequency Prior:* To locate the salient regions in multimodal remote sensing images and ensure the repeatability of keypoints, we adopt a detection loss with a high-frequency prior for network optimization, which consists of peaking loss and high-frequency consistency loss. To encourage the detection scores of keypoints peak locally, our PPDet is primarily regularized using the peaking loss function. Given the detection score map $S^{opt}$ and $S^{oth}$, the peaking loss $\mathcal{L}_{s-p}$ can be computed as

$$\begin{aligned}\mathcal{L}_{s-p} = &\left(AP\left(S^{opt}\right)+1\right)^2 + \left(1 - MP\left(S^{opt}\right)\right)^2 \\ &+ \left(AP\left(S^{oth}\right)+1\right)^2 + \left(1 - MP\left(S^{oth}\right)\right)^2\end{aligned} \tag{7}$$

where $AP$ and $MP$ represent the average pooling and the max pooling, respectively.

Suffering from severe NRDs, it is still difficult to locate the salient regions and extract more repeatable points for heterogeneous image matching. According to [21], phase information of images is much more invariant to illumination and contrast changes, which is important for cross-modal image perception. To alleviate the differences between detection score maps and identify salient regions with highly consistent local phase information, we produce the high-frequency prior mask $P_{hfp}$ to improve the distribution of keypoints utilizing PIs. Since detection scores of keypoints lying on corresponding areas are expected to be more similar and larger, the high-frequency consistency loss $\mathcal{L}_{hfc}$ can be formulated as

$$P_{hfp} = Norm_{+}\left(AP\left(MP\left(\nabla^2\left(PI\right)\right)\right)\right) \tag{8}$$

$$\mathcal{L}_{hfc} = \left(P_{hfp}^{opt}\left(1 - S^{opt}\right)\right)^2 + \left(P_{hfp}^{oth}\left(1 - S^{oth}\right)\right)^2 \tag{9}$$

where $\nabla^2$ represents the second-order spatial gradient operator to further extract precisely the structural cues from PIs that define reliable keypoint locations, and $Norm_{+}$ means the normalization with ReLU to ensure the high-frequency prior mask is nonnegative. The combined use of maximum pooling and average pooling enhances the saliency of common high-frequency regions while suppressing the high-frequency noise. So the high-frequency consistency loss can push the model to assign high scores to keypoints in high-frequency consistent regions, which can force it to detect more repeatable and reliable keypoints.

*2) Spatial-Frequency Description Loss:* Following [55], we utilize the angular distance $\theta^{cross}$ to measure the similarity of the corresponding descriptors $(d_i^{opt}, d_i^{oth})$ to strike a balance between matching pairs and nonmatching pairs. It can be formulated as

$$\theta^{cross} = arccos\left(\left(d_i^{opt}\right)^T d_i^{oth}\right). \tag{10}$$

Given the cross-modal descriptors set $\{d_i^{opt}, d_i^{oth}\}$ $(i = 1, 2, 3, \ldots, K)$, we can compute the angular similarity matrix $M_{cross} \in \mathbb{R}^{K\times K}$. To prevent the feature similarity across different positions from becoming overly consistent, leading to failure in identifying the correct matching position, we also introduce the peak loss $\mathcal{L}_{d-p}$ for feature similarity discrimination, which can be formulated as

$$\mathcal{L}_{d-p} = \left(2\pi - \left(max\left(M_{cross}\right) - mean\left(M_{cross}\right)\right)\right)^2. \tag{11}$$

Similarly, we can compute the similarity matrix of intra-modal descriptors $M_{intra}$. To mine more hard negative samples, we combine spatial-domain and frequency-domain information to constrain the sampling area. In particular, we utilize the high-frequency prior mask $P_{hfp}$, the similarity matrix $M_{cross}$ and $M_{intra}$ to select *Top-N* pixels with the highest similarity in the same high-frequency regions. The spatial-frequency loss $\mathcal{L}_{sf}$ can be computed as

$$\begin{aligned}\mathcal{L}_{sf} = &\Bigg[\left(\pi - \frac{1}{N}\sum_{k=1}^{N}\left(\text{Top}\,N\left(M_{\text{cross}}\right)\right)\right)^2 \\ &+ \left(\pi - \frac{1}{N}\sum_{k=1}^{N}\left(\text{Top}\,N\left(M_{\text{intra}}\right)\right)\right)^2 \\ &+ \left(\max\left(M_{\text{cross}}\right)\right)^2\Bigg]^2\end{aligned} \tag{12}$$

where $N$ is set to 32 for the best experimental performance.

The final optimization loss for P2D2 can be represented as

$$W_{det} \triangleq S^{opt} S^{oth} \tag{13}$$

$$W_{des} \triangleq Norm_{+}\left(1 - \frac{M_{cross}}{mean\left(M_{cross}\right)}\right) \tag{14}$$

$$\mathcal{L} = W_{des}\left(\mathcal{L}_{s-p} + \mathcal{L}_{hfc}\right) + W_{det}\left(\mathcal{L}_{d-p} + \mathcal{L}_{sf}\right) \tag{15}$$

where $\triangleq$ denotes the stop-gradient operation, $W_{det}$ and $W_{des}$ are the weights to mutually couple detection and description. In particular, $W_{det}$ prioritizes the optimization of descriptors with relatively higher detection scores while $W_{des}$ encourages more reliable descriptors for better probability detection. These weights are both dynamically normalized and detached from back propagation, which enables stable joint training of detection and description, while ensuring they are not disturbed by the gradients of these weights.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

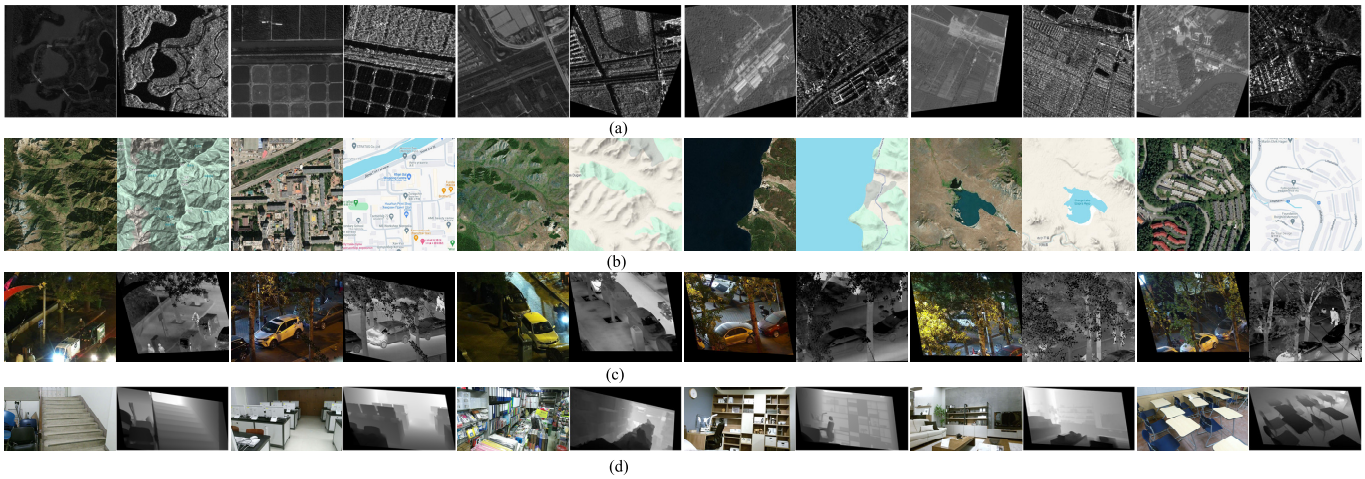



Fig. 4. Some examples of our collected multimodal remote sensing image dataset. (a) OPT-SAR dataset. (b) OPT-MAP dataset. (c) OPT-NTIR dataset. (d) OPT-DEP dataset.

TABLE I
BASIC INFORMATION ABOUT OUR DATASET

| Category | Description | Dataset size | | Image size | Resolution(m) | Perspective distortion | Rotation change | Scale change |
|---|---|---|---|---|---|---|---|---|
| | | Trian | Test | | | | | |
| OPT-SAR | Satellite remote sens. image | 2000 | 400 | $512 \times 512$ | 1 | ✓ | ✓ | ✓ |
| OPT-MAP | Satellite remote sens. image | 500 | 100 | $512 \times 512$ | 3-500 | ✓ | ✓ | ✓ |
| OPT-NTIR | Outdoor image at nighttime | 500 | 100 | $640 \times 512$ | Unknown | ✓ | ✓ | ✓ |
| OPT-DEP | Indoor close range image | 1500 | 300 | $512 \times 288$ | Unknown | ✓ | ✓ | ✓ |

## IV. EXPERIMENTS

In this section, a multimodal remote sensing image dataset comprising four representative image categories is introduced for the evaluation in both quantitative and qualitative metrics. Then, we will present the settings, results, and analysis of experiments based on this dataset.

### A. Experimental Datasets

Four types of multimodal remote sensing images are collected for evaluation, including OPT-SAR, OPT-MAP, OPT-NTIR, and OPT-DEP. Some examples in our collected multimodal remote sensing image dataset are shown in Fig. 4 and detailed information on the dataset is shown in Table I.

*1) OPT-SAR:* The OPT-SAR dataset is created based on the OS-DATASET [56]. The OS-DATASET consists of 2673 registered optical-SAR image pairs in the uniform size of $512 \times 512$ pixels, which span across cities around the world and cover various scenes. The optical images are collected from the Google Earth platform, and SAR images are obtained by the Chinese GaoFen-3 (GF-3) with the same resolution of 1 m. There are 2000 pairs of optical and SAR images for training and 400 pairs for testing. To evaluate the geometric invariance of the proposed method, we randomly generate homography transforms with a perspective distortion factor $p \in [0, 0.2]$, a rotation angle $r \in [-10°, 10°]$, and a scale factor $s \in [0.8, 1.0]$ and apply them to the target images to generate our test set.

*2) OPT-MAP:* The dataset collected by Wu et al. [57] is used to produce our OPT-MAP dataset, in which the electronic navigation maps are obtained using the Google Maps API, and their corresponding optical images are downloaded from the Google Earth Engine (GEE) platform with resolutions ranging from 3 to 500 m. The OPT-MAP dataset comprises 600 optical-map image pairs with pixel dimensions of $512 \times 512$. A total of 500 image pairs are used for training and 100 pairs for testing. In the coarse annotation stage, we employ the RANSAC algorithm to estimate an initial homography matrix, with a strict reprojection error threshold of 3 pixels. In the fine annotation stage, an automated template matching refinement is iteratively applied to ensure that the final reprojection error is controlled within 1 pixel. All image pairs are then validated via the a-contrario method [58] and manually rechecked.

*3) OPT-NTIR:* To assess the performance of matching in low-light conditions, we collect 600 pairs of optical and IR images taken at nighttime from the LLVIP dataset [59], which are strictly aligned in both time and space. The optical images and their corresponding IR images are both resized to $640 \times 512$ pixels, and the dataset is divided into a training set of 500 image pairs and a test set of 100 image pairs. Consistent with the OPT-SAR dataset, we also add random perspective distortion, rotation, and scale change to the target images.

*4) OPT-DEP:* To investigate the generalization performance on indoor image matching, we produce the OPT-DEP dataset based on the DIML/CVL RGB-D dataset [60], which consists of optical images and their corresponding depth maps captured by Microsoft Kinect v2. This dataset contains 2000 pairs of optical-depth image pairs in the size of $512 \times 288$ pixels and covers 26 representative indoor scenes such as library, laboratory, and restaurant. 1500 image pairs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

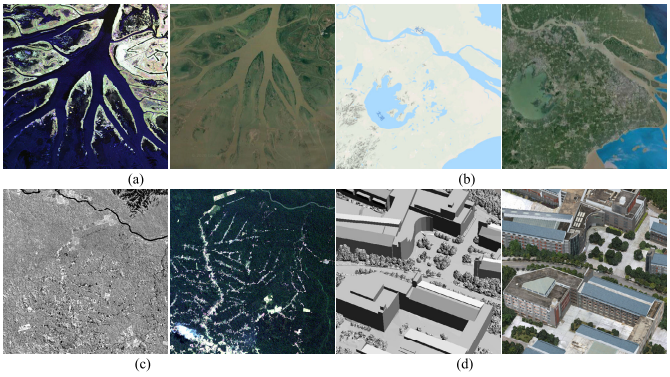



Fig. 5. Examples of test datasets for cross-dataset generalization evaluation. (a) SAR-Opti. (b) MAP-Opti. (c) IR-Opti. (d) Depth-Opti.

are for training, and 300 image pairs are preprocessed in the same way as the OPT-SAR dataset for testing.

In addition, we use four publicly available datasets [1] for cross-dataset generalization evaluation, including SAR-Optical, MAP-Optical, IR-Optical, and Depth-Optical. As shown in Fig. 5, these test datasets exhibit substantial visual discrepancies compared to our collected dataset, particularly the SAR-Optical and Depth-Optical datasets.

### B. Evaluation Metrics

To validate the effectiveness of detection, we report the number of corresponding keypoints (NCK) and the repeatable rate (RR), i.e., the ratio of the NCK located in the common-observation area to the total number of all detected keypoints. We use six metrics for the quantitative evaluation of description, including the number of correct matches (NCM), matching success rate (MSR), precision (Pre), recall (Rec), $F1$-score ($F1$), and root-mean-square error (RMSE) of correct matches.

Given the ground truth transformation $H_{gt}$, the definition of $NCM$ is

$$NCM = \sum_{i}^{NCK} \left[ \left| \text{ref}\,(x_i, y_i) - H_{gt} * \text{sen}\,(x_i, y_i) \right| < \epsilon \right] \tag{16}$$

where $\text{ref}(x, y)$ represents a point on the reference image and $\text{sen}(x, y)$ is the corresponding feature point of $\text{ref}(x, y)$ on the sensed image. $\epsilon$ is the pixel threshold for determining whether the match is correct.

Once the $NCM$ of the image pair is larger than 10, we determine that this image pair has been successfully matched. So the $MSR$ is calculated by

$$MSR = \frac{\text{Number of successfully matched pairs}}{\text{Size of test dataset}}. \tag{17}$$

And we calculate the $RMSE$ as

$$RMSE = \sqrt{\frac{1}{NCM} \sum_{i=1}^{NCM} \left[ \left(x_i - x_i'\right)^2 + \left(y_i - y_i'\right)^2 \right]} \tag{18}$$

where $(x_i, y_i)$ is a point of correct matches and $(x_i', y_i')$ represents the corresponding point of it after the true matrix transformation. In cases where the NCM of image pairs is less than 10, the RMSE values are uniformly set to 20 pixels.

### C. Implement Details

The proposed P2D2 is implemented using the PyTorch framework. The constant amplitude $A_{const}$ of PIs is uniformly set at 5e4 on four datasets. The network of P2D2 is trained for 50 epochs with a batch size of 2 on an NVIDIA GeForce RTX 4090 GPU. During the training phase, the Adam optimizer is introduced for network optimization, with the learning rate update strategy comprising an initial warm-up phase followed by a cosine annealing schedule. In particular, the learning rate linearly increases from 0 to 1e-3 during the warm-up phase, then follows a decay schedule, reaching zero at the final epoch. However, the final checkpoint saved during the training process is employed for evaluation. The experiments were randomly executed ten times, and statistical significance testing was performed using a paired sample t-test.

### D. Comparison Experiments

To demonstrate the efficacy of the proposed method, the proposed P2D2 is compared with seven state-of-the-art methods, i.e., OS-SIFT [17], RIFT [22], LNIFT [23], SRIF [26], WSSF [42], ReDFeat [30], and AttFeat [35]. All implementations of the comparison methods employed in the experiments were either provided by the authors or sourced from publicly available software. In particular, we uniformly set the maximum number of extracted key points to 5000 for all compared methods and choose the brute force matching with the bidirectional nearest neighbor searching as the matching strategy.

The quantitative performance of feature matching for the aforementioned eight methods under the threshold of 3 pixels is quantified in Table II. As can be seen, the proposed P2D2 achieves the state-of-the-art NCK and RR on the four datasets, which demonstrates that the proposed method can detect more reliable and repeatable keypoints. In particular, P2D2 generates over 200 more corresponding keypoints, and the RR of keypoints is nearly 4.6% higher than the second-place AttFeat on the OPT-MAP dataset (paired t-test, $p < 0.01$). Besides, the proposed method achieves 3.0% higher RR on the OPT-NTIR dataset and approximately 800 additional corresponding keypoints are obtained on the OPT-DEP dataset (paired t-test, $p < 0.01$). Due to the noise interference of SAR images, P2D2 is slightly better than the detection performance of the other methods on the OPT-SAR dataset. In terms of description, the MSR of the proposed method on all datasets has reached 100%, which verifies the robustness of our proposed DSIADes. Moreover, P2D2 achieves the highest NCM, Pre, $F1$, and the lowest RMSE on the first three datasets. In particular, the NCM of the proposed method is 38.7%, 19.1%, and 4.2% higher than the suboptimal result on OPT-SAR, OPT-MAP, and OPT-NTIR datasets, respectively (paired t-test, $p < 0.01$). However, due to the lack of textural information of depth maps, DSIADes cannot extract effective structural information from the PIs, leading to the NCM, Pre, Rec, and $F1$ of the proposed method being marginally lower than the optimal one on the OPT-DEP dataset.

Furthermore, the RR of detection and the NCM of description are selected as the representative metrics and calculated at discrete threshold intervals. As we can see in Fig. 6,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE EVALUATION RESULTS ON FOUR DATASETS. THE 1ST, 2ND, AND 3RD-BEST METHODS ARE HIGHLIGHTED. THE PIXEL THRESHOLD IS THREE

| Category | Method | Detection | | Description | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | NCK ↑ | RR (%) ↑ | NCM ↑ | MSR (%) ↑ | Pre (%) ↑ | Rec (%) ↑ | F1 (%) ↑ | RMSE (px) ↓ |
| OPT-SAR | OS-SIFT | 468.80 | 20.143 | 6.3825 | 21.50 | 1.38 | 0.90 | 1.08 | 16.772 |
| | RIFT | 2502.3 | 50.047 | 41.222 | 76.25 | 1.56 | 4.53 | 2.30 | 6.7475 |
| | LNIFT | 1341.5 | 26.831 | 73.977 | 96.25 | 5.10 | 3.52 | 4.12 | 2.7605 |
| | SRIF | 1091.2 | 21.824 | 189.87 | 99.00 | 7.25 | 4.59 | 5.61 | 2.2341 |
| | WSSF | 1529.2 | 30.584 | 61.142 | 91.25 | 3.83 | 6.26 | 4.70 | 3.6530 |
| | ReDFeat | 2862.1 | 57.243 | 191.68 | 98.50 | 6.48 | 4.13 | 5.04 | 2.3201 |
| | AttFeat | 2865.7 | 57.314 | 45.475 | 87.00 | 1.50 | 0.97 | 1.18 | 4.3838 |
| | P2D2 | 2871.1 | 57.423 | 265.87 | 100.0 | 15.54 | 14.32 | 14.62 | 2.2305 |
| OPT-MAP | OS-SIFT | 296.97 | 23.501 | 10.340 | 34.00 | 3.35 | 2.41 | 2.72 | 13.901 |
| | RIFT | 2244.4 | 45.084 | 94.390 | 84.00 | 4.22 | 9.55 | 5.76 | 4.8989 |
| | LNIFT | 758.27 | 15.165 | 54.930 | 91.00 | 6.45 | 4.36 | 5.10 | 3.6932 |
| | SRIF | 650.42 | 13.008 | 139.31 | 96.00 | 18.77 | 14.74 | 16.07 | 2.7878 |
| | WSSF | 1455.9 | 29.238 | 286.23 | 93.00 | 18.15 | 23.51 | 20.26 | 3.3074 |
| | ReDFeat | 2655.7 | 53.114 | 779.08 | 98.00 | 29.72 | 17.99 | 22.36 | 2.3185 |
| | AttFeat | 2667.8 | 53.356 | 156.14 | 80.00 | 5.41 | 3.17 | 3.99 | 5.5573 |
| | P2D2 | 2898.4 | 57.968 | 927.86 | 100.0 | 30.17 | 17.91 | 22.48 | 1.9357 |
| OPT-NTIR | OS-SIFT | 569.56 | 21.146 | 11.260 | 57.00 | 2.30 | 1.80 | 1.98 | 11.912 |
| | RIFT | 2477.3 | 49.547 | 89.610 | 95.00 | 3.57 | 7.82 | 4.88 | 3.1305 |
| | LNIFT | 1023.0 | 20.460 | 60.970 | 93.00 | 5.86 | 4.22 | 4.88 | 3.3485 |
| | SRIF | 863.97 | 17.279 | 155.95 | 99.00 | 17.24 | 10.38 | 12.94 | 2.2481 |
| | WSSF | 1581.6 | 31.632 | 97.900 | 99.00 | 6.18 | 10.59 | 7.78 | 2.2188 |
| | ReDFeat | 2828.5 | 56.570 | 491.95 | 100.0 | 17.52 | 10.02 | 12.73 | 1.9936 |
| | AttFeat | 2854.9 | 57.099 | 265.27 | 100.0 | 8.96 | 5.40 | 6.73 | 1.9811 |
| | P2D2 | 3000.0 | 60.000 | 512.66 | 100.0 | 17.60 | 14.62 | 15.83 | 1.9802 |
| OPT-DEP | OS-SIFT | 149.06 | 21.101 | 9.7166 | 40.00 | 6.50 | 4.90 | 5.43 | 13.517 |
| | RIFT | 2646.5 | 75.690 | 193.38 | 96.33 | 7.15 | 27.12 | 11.24 | 2.7206 |
| | LNIFT | 1826.3 | 36.527 | 186.77 | 100.0 | 9.85 | 13.20 | 11.20 | 2.0071 |
| | SRIF | 1658.5 | 33.171 | 247.95 | 100.0 | 14.10 | 27.32 | 18.44 | 1.9660 |
| | WSSF | 1795.6 | 39.814 | 70.933 | 98.33 | 3.86 | 19.68 | 6.42 | 2.2204 |
| | ReDFeat | 3278.4 | 100.00 | 1188.4 | 100.0 | 36.35 | 35.40 | 35.78 | 1.7442 |
| | AttFeat | 3510.9 | 100.00 | 686.50 | 100.0 | 19.67 | 19.43 | 19.49 | 1.6533 |
| | P2D2 | 4270.8 | 100.00 | 1154.5 | 100.0 | 26.84 | 27.33 | 27.08 | 1.9526 |

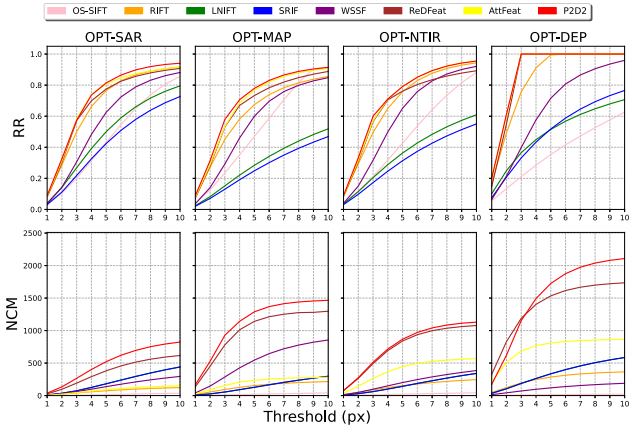



Fig. 6. Feature matching performance of eight methods on four datasets. RR of detected points and the NCM points are drawn in curves with pixel thresholds from 1 to 10.

P2D2 demonstrates the best performance of detection and description at all thresholds on the first three datasets and at pixel thresholds from 4 to 10 on the OPT-DEP dataset. The aggregate performance of OS-SIFT, RIFT, LNIFT, SRIF, and WSSF is weaker than that of the other three deep learning-based methods due to the lack of training on datasets. RIFT and LNIFT lack scale invariance while being susceptible to viewpoint variations. OS-SIFT, SRIF, WSSF, and AttFeat are all constrained by the robustness of their respective descriptors, exhibiting favorable performance solely on specific datasets. In contrast, ReDFeat still has scope for improvement owing to the lack of a more interpretable detection.

As for detection, the RR curves of P2D2 override other methods on all datasets. Apart from P2D2, RIFT, and AttFeat, respectively, attain the highest RR among manual and learning-based methods under different thresholds and across diverse data scenarios. This is mainly because RIFT detects both corner points and edge points on the phase congruency map, while AttFeat utilizes a parallelized attention structure to focus on important spatial locations. P2D2 takes advantage of both frequency-domain and spatial-domain analyses by producing PIs at a reasonable computational cost and mining distinctive information with deep learning techniques, which considers both the repeatability and interpretability of corresponding keypoints. As for NCM, P2D2 gains large margins on OPT-SAR and OPT-MAP subsets and is marginally superior to other methods on OPT-NTIR subset at all thresholds. Besides, P2D2 only slightly lags behind at thresholds from 1 to 3 but significantly leads at thresholds from 4 to 10 on the OPT-DEP subset. These quantitative results on four datasets confirm the superiority of the proposed method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

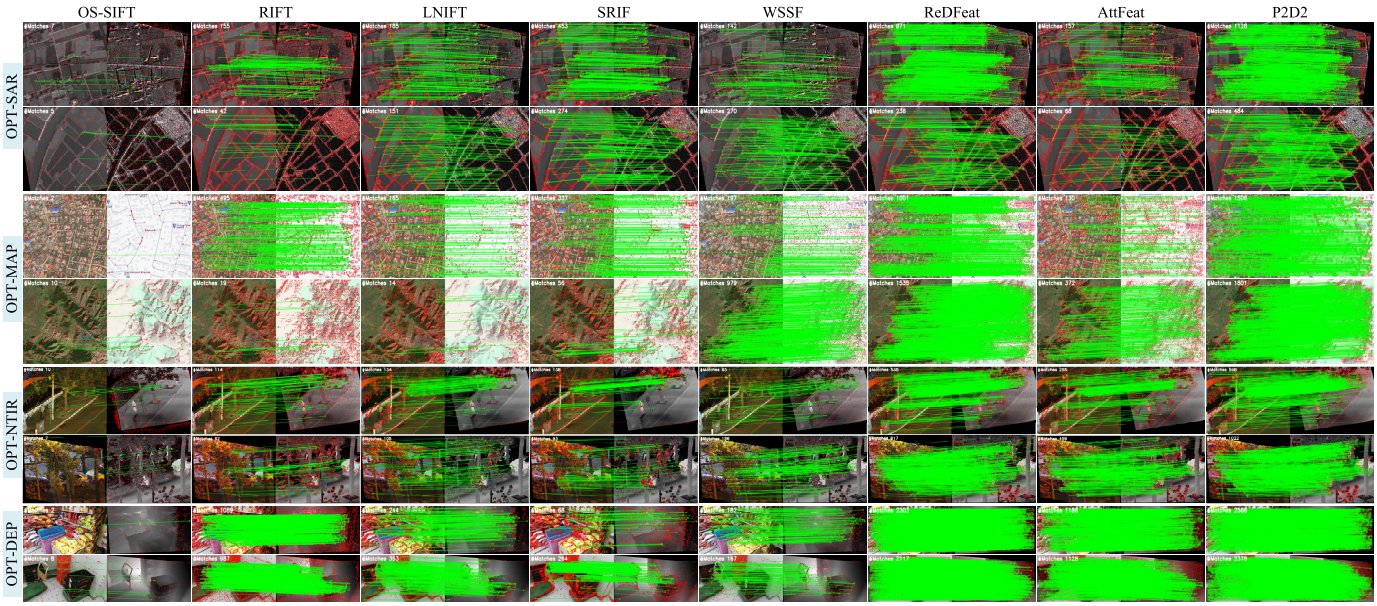



Fig. 7. Visualization of matching performance on our collected multimodal remote sensing image dataset.

TABLE III

MATCHING PERFORMANCE ON PUBLIC MULTIMODAL REMOTE SENSING IMAGE DATASETS WITHOUT FINE-TUNING. THE BEST AND SECOND RESULTS ARE, RESPECTIVELY, MARKED AS BOLD AND UNDERLINE

| Category | Method | Detection | | Description | | |
|---|---|---|---|---|---|---|
| | | NCK ↑ | RR (%) ↑ | NCM ↑ | MSR (%) ↑ | RMSE (px) ↓ |
| SAR-Opti. | ReDFeat | 2379.6 | 47.593 | 524.83 | 100.0 | 2.0039 |
| | AttFeat | 2425.6 | 48.513 | 212.50 | 100.0 | 2.0105 |
| | P2D2 | **2567.3** | **51.346** | **549.33** | **100.0** | **1.9347** |
| MAP-Opti. | ReDFeat | 1969.7 | 39.394 | 98.000 | 85.71 | 4.5944 |
| | AttFeat | 2096.4 | 41.928 | 8.7142 | 42.86 | 12.229 |
| | P2D2 | **2271.1** | **45.422** | **253.71** | **100.0** | **2.0583** |
| IR-Opti. | ReDFeat | 2386.0 | 47.720 | 195.75 | 100.0 | **1.9559** |
| | AttFeat | 2401.7 | 48.034 | 181.00 | 75.00 | 6.3902 |
| | P2D2 | **2490.7** | **49.815** | **343.75** | **100.0** | 2.0473 |
| Depth-Opti. | ReDFeat | 2432.3 | 48.646 | **821.75** | 100.0 | 1.8466 |
| | AttFeat | 2371.2 | 48.139 | 216.87 | 100.0 | **1.7384** |
| | P2D2 | **2435.7** | **48.714** | 796.12 | **100.0** | 1.9874 |

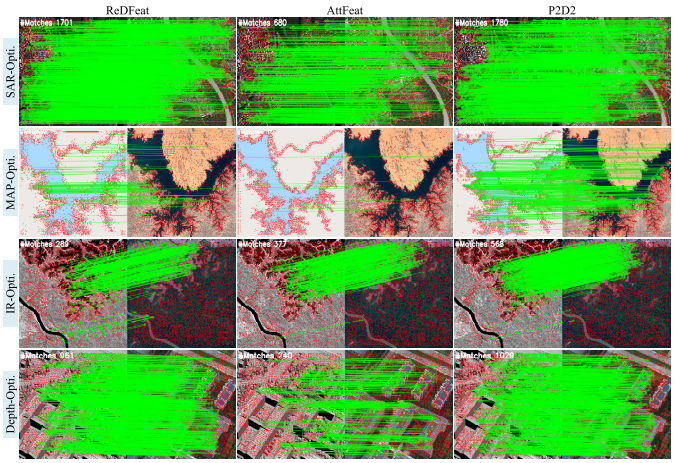



Fig. 8. Visualization of cross-dataset generalization evaluation.

To evaluate the cross-dataset generalization of P2D2, we additionally test it on publicly available datasets without any fine-tuning. As is shown in Table III, P2D2 once again achieves the MSR of 100% across all datasets, demonstrating its robustness to substantial visual discrepancies. Compared with state-of-the-art learning-based methods, P2D2 consistently provides superior feature detection and description performance. Notably, in the challenging MAP-Optical scenario, P2D2 improves the RR by 3.5% over AttFeat and its NCM is approximately 2.6 times that of ReDFeat (paired t-test, $p < 0.01$). On the Depth-Optical dataset, the NCM of P2D2 is only slightly below the optimal, due to low-texture regions. These results collectively demonstrate the effectiveness and stable generalization capability of the proposed P2D2.

### E. Visualization

To facilitate an intuitive comparison of the performance of feature detection and description, visual comparisons of typical image pairs across four datasets are presented in Fig. 7. A total of 5000 keypoints are detected by each method and denoted by red crosshairs. The correct matches under the threshold of 3 pixels are represented as green lines, and the NCM is displayed in the upper left corner of each image pair. It is worth noting that some methods can only extract a small number of keypoints (typically well below 5000) in cross-modal image pairs, especially in the OPT-MAP dataset with extremely severe nonlinear NRDs and the OPT-DEP dataset suffering from the lack of textural information.

As shown, the proposed PPDet mainly extracts keypoints in salient regions, which usually contain more high-level image content and semantic information. Though the distribution of keypoints detected by OS-SIFT, RIFT, WSSF, ReDFeat, and AttFeat may be more uniform, a considerable number of keypoints are lying on smooth areas without texture, which may hinder subsequent matching due to the lack of discriminative information. Benefiting from the high-frequency prior provided by PIs, keypoints extracted by PPDet basically fall in corners, edges, or blobs with highly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

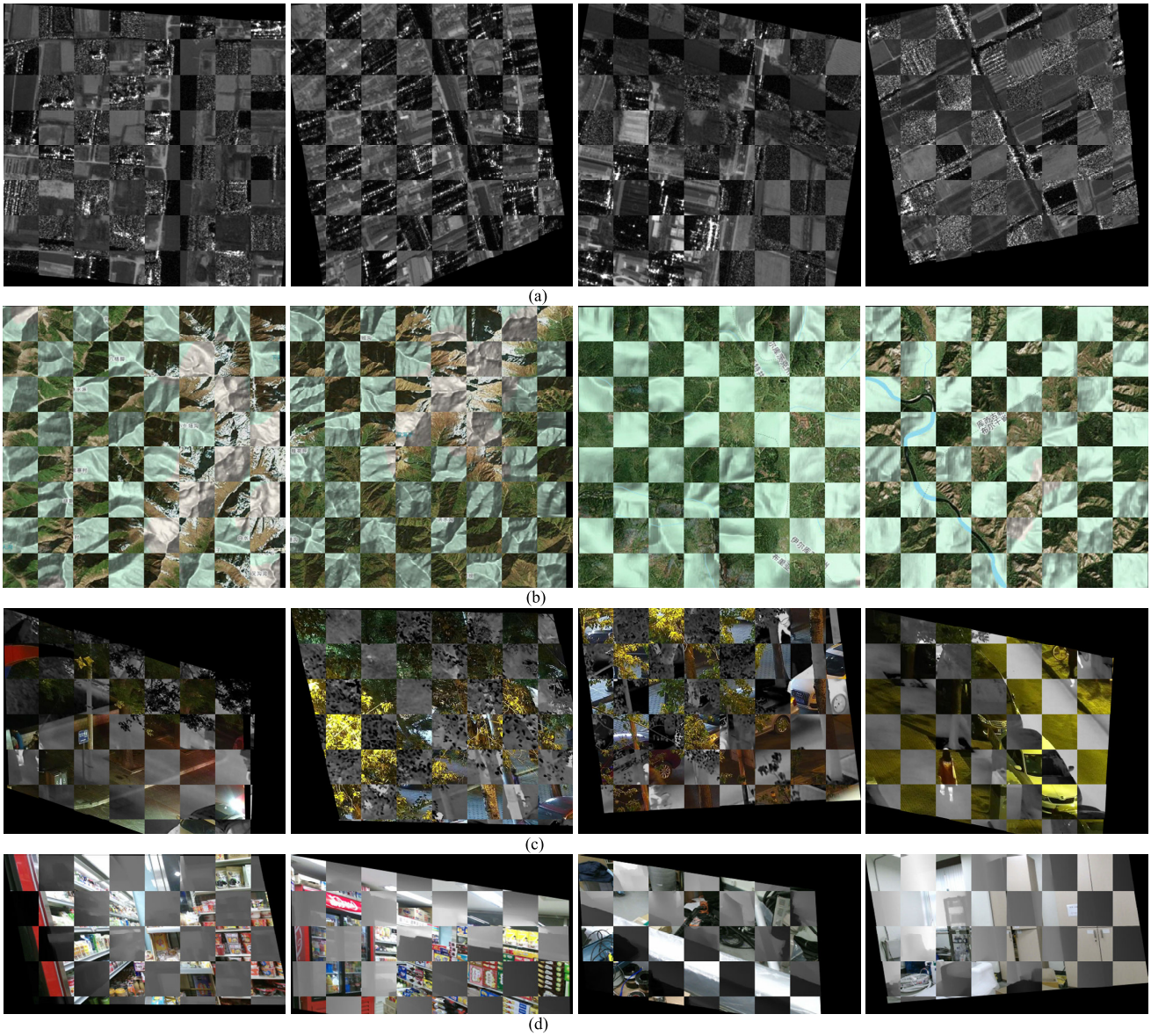



Fig. 9. Visualization of image registration performance on our collected multimodal remote sensing image dataset. (a) OPT-SAR. (b) OPT-MAP. (c) OPT-NTIR. (d) OPT-DEP.

consistent local phase information, which makes our detection more interpretable. Based on more repeatable detected points, the proposed DSIADes achieves desirable performance in feature matching on four datasets. Whereas other methods can only generate a limited number of matches when confronted with severe NRDs and geometric differences, P2D2 still establishes a considerable number of correspondences in such cases, which indicates the great potential of combining spatial information with frequency-domain information for multimodal remote sensing image matching. Fig. 8 shows several feature matching results of three learning-based methods on public datasets without fine-tuning. As observed from the results, P2D2 prevents points from being distributed within featureless regions, thereby yielding superior matching performance.

We also provide several visualization results of image registration of the proposed method on four datasets. As illustrated in Fig. 9, we present the checkboard images and show the remarkable smoothness of the intersections between optical and other modal images, thereby highlighting the substantial benefits of the proposed method for multimodal remote sensing image registration.

### F. Ablation Study

*1) Impact of $A_{const}$:* In the PI extractor, $A_{const}$ is introduced to control the numerical scale of the phase-only reconstruction. To investigate its impact, we train and evaluate P2D2 with different values of $A_{const}$ on four datasets. Fig. 10 reports the RR and NCM of P2D2 using 5000 keypoints under varying amplitudes. The results show that an amplitude of 5e4 consistently provides strong and near-optimal feature matching performance across all datasets. From a numerical signal processing perspective, overly small amplitudes lead to low-contrast PIs, while excessively large values amplify high-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

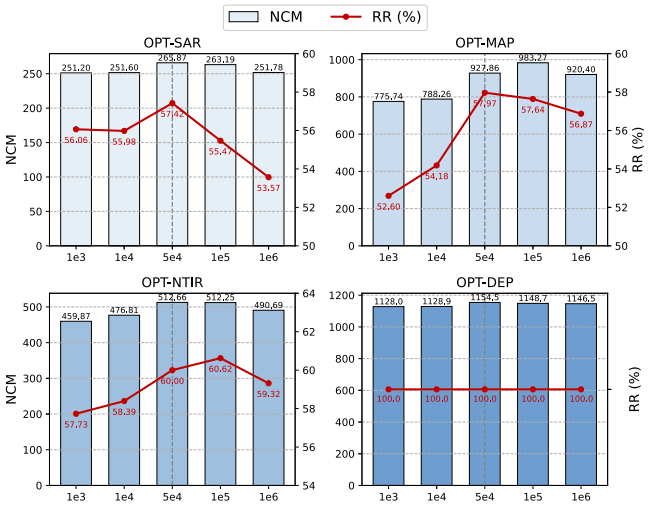



Fig. 10. Ablation study of the amplitude parameter on four datasets.

TABLE IV
COMPONENTWISE ABLATION RESULTS ON OPT-MAP DATASET

| Method | Detection | | Description | |
|---|---|---|---|---|
| | NCK ↑ | RR (%) ↑ | NCM ↑ | RMSE (px) ↓ |
| Two-branch network | 2532.8 | 50.657 | 813.57 | 1.9809 |
| + Phase perception branch | 2639.0 | 52.781 | 868.49 | 1.9855 |
| + High-frequency prior | 2737.2 | 54.744 | 879.72 | 1.9781 |
| + DSIA module | 2792.2 | 55.844 | 907.41 | 1.9467 |
| + Spatial-Frequency loss (P2D2) | **2898.4** | **57.968** | **927.86** | **1.9357** |

frequency noise. Based on this analysis, $A_{const}$ is fixed to 5e4 to ensure stable and consistent performance.

*2) Proposed Components:* To evaluate the efficacy of key components in the proposed method, we conducted ablation experiments based on the OPT-MAP dataset. Using the two-branch network as the baseline, we systematically evaluate the performance gains achieved by progressively integrating the individual components of P2D2. However, the experimental results of the componentwise comparison are presented in Table IV. The bold values represent the best score columnwise.

As can be seen, when the phase-enhanced branch is inserted into the baseline, the number of corresponding keypoints (NCK) increases by 4.19% and the NCM increases by 6.75%, which indicates that frequency-domain information provided by PIs enriches the feature representation. Given that a high-frequency prior is introduced, the RR of keypoints increases by 1.96%, which demonstrates that more keypoints located in high-frequency regions are conducive to detection. Furthermore, with the diverse structural information augmented (DSIA) module, the proposed descriptor exhibits stronger discriminability and robustness, contributing to a 3.15% increase in the metric NCM and a 1.59% decrease in the metric RMSE. Finally, as the spatial-frequency loss is incorporated for network optimization, the proposed P2D2 achieves the best performance of both detection and description.

### G. Running Time Analysis

To evaluate the computational efficiency of the proposed P2D2, we report the average running time of P2D2 and four other handcrafted methods on four datasets with different image sizes. All experiments are run on a PC with an Intel[1] Core[2] i9-14900K CPU and an NVIDIA RTX A6000 GPU.

TABLE V
COMPARISONS ON RUNNING TIME (SECONDS). THE AVERAGE SIZES OF IMAGES ARE GIVEN

| | Category | OPT-SAR | OPT-MAP | OPT-NTIR | OPT-DEP | Average |
|---|---|---|---|---|---|---|
| | Image size (pixel) | 512 × 512 | 512 × 512 | 640 × 512 | 512 × 288 | |
| Method | OS-SIFT | 3.279 | 3.760 | 4.016 | 0.941 | 2.999 |
| | RIFT | 14.27 | 13.65 | 19.20 | 10.60 | 14.43 |
| | LNIFT | 1.257 | 1.475 | 1.285 | 0.874 | 1.222 |
| | WSSF | 11.88 | 10.81 | 15.04 | 8.839 | 11.64 |
| | P2D2 | 1.647 | 1.654 | 1.668 | 1.026 | 1.499 |

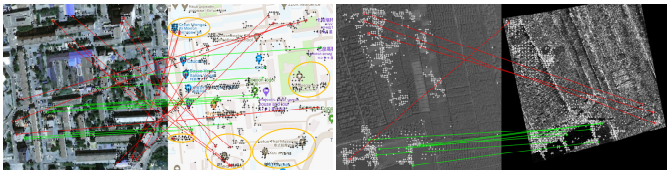


Fig. 11. Visualization of some failure cases.

As we can see in Table V, the proposed method runs much faster than RIFT and WSSF, with its average runtime surpassed only by LNIFT. For instance, P2D2 is about two times faster than OS-SIFT, 12 times faster than RIFT, and nine times faster than WSSF on the OPT-NTIR dataset. It is worth noting that the proposed method extracts many more corresponding keypoints than the four other handcrafted methods, and the majority of our running time is spent on the bidirectional nearest neighbor searching, which benefits from our simple yet effective PI extractor to explore the shared frequency-domain information. In contrast, RIFT suffers from the high computational complexity of constructing the maximum index map, while WSSF consumes too much time for the calculation of the structural salience map to build local descriptors. Furthermore, the running time of P2D2 is less sensitive to variations in image size. Considering the improvements of both detection and description, it is believed that P2D2 achieves the performance-efficiency balance for multimodal remote sensing image matching.

### H. Failure Case Analysis

Despite achieving competitive performance across various scenarios, our method remains constrained by certain limitations in specific cases. As illustrated in Fig. 11, keypoints in the map are mainly concentrated on the landmarks and their corresponding texts, which prevents PPDet from recognizing meaningful feature points and leads to poor keypoint repeatability. Besides, noise interference and lack of structural information decrease the robustness of our descriptor and affect matching accuracy. Resolving these aforementioned issues will be incorporated into our future work.

## V. CONCLUSION

Due to the severe NRDs, it is still a challenging problem to achieve accurate and robust multimodal remote sensing

[1]Registered trademark.
[2]Trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

image matching. Existing methods predominantly focus on mining modal-invariant information within either the spatial or frequency domain, while lacking concurrent exploration of cross-domain information integration. This article proposes a joint feature detection and description framework named P2D2, which integrates spatial-domain and frequency-domain information to improve the feature extraction and description capabilities via phase cue feature learning. To efficiently explore the useful frequency-domain information, we apply the PI extractor to produce the PIs, which contain high-frequency phase information conducive to both feature detection and description. Then, we utilize an unshared lightweight tri-branch encoder to mitigate the nonlinear radiometric differences between multimodal images. To extract modal-invariant keypoints at multiple levels, we design the PPDet based on an attention gate mechanism to locate the salient regions in cross-modal images. However, we innovatively introduce the high-frequency prior generated by PIs for detection optimization, which is beneficial to locating the salient regions in cross-modal images and extracting more repeatable and interpretable keypoints. In addition, we design the DSIADes to aggregate diverse surrounding structural information of PIs from different levels and different receptive fields and learn the cross-modality consistent feature representation. Experimental results indicate that the proposed P2D2 can achieve improved performance of both detection and description over other spatial-domain methods and frequency-domain methods on multimodal remote sensing image matching. In future work, we will explore more robust feature learning methods for multimodal remote sensing images lacking textural and structural information.

## REFERENCES


[1] X. Jiang, J. Ma, G. Xiao, Z. Shao, and X. Guo, "A review of multimodal image matching: Methods and applications," *Inf. Fusion*, vol. 73, pp. 22–71, Sep. 2021.

[2] H. Zhang, H. Xu, X. Tian, J. Jiang, and J. Ma, "Image fusion meets deep learning: A survey and perspective," *Inf. Fusion*, vol. 76, pp. 323–336, Dec. 2021.

[3] H. Xu, J. Yuan, and J. Ma, "MURF: Mutually reinforcing multi-modal image registration and fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12148–12166, Oct. 2023.

[4] J. Chen, D. Hou, C. He, Y. Liu, Y. Guo, and B. Yang, "Change detection with cross-domain remote sensing images: A systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11563–11582, 2024.

[5] R. Zhou et al., "A unified deep learning network for remote sensing image registration and change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5101216.

[6] S. Łukasik, S. Szott, and M. Leszczuk, "Multimodal image-based indoor localization with machine learning—A systematic review," *Sensors*, vol. 24, no. 18, p. 6051, Sep. 2024.

[7] Z. Wang et al., "Sequence matching for image-based UAV-to-satellite geolocalization," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607815.

[8] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.

[9] H. Bay, T. Tuytelaars, and L. Van Gool, "SURF: Speeded-up robust features," in *Proc. 9th Eur. Conf. Comput. Vis.*, vol. 3951, 2006, pp. 404–417.

[10] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 2564–2571.

[11] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-supervised interest point detection and description," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 224–236.

[12] J. Revaud, C. R. D. Souza, M. Humenberger, and P. Weinzaepfel, "R2D2: Reliable and repeatable detector and descriptor," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 12405–12415.

[13] X. Zhao, X. Wu, J. Miao, W. Chen, P. C. Y. Chen, and Z. Li, "ALIKE: Accurate and lightweight keypoint detection and descriptor extraction," *IEEE Trans. Multimedia*, vol. 25, pp. 3101–3112, 2022.

[14] P. Gleize, W. Wang, and M. Feiszli, "SiLK: Simple learned keypoints," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 22442–22451.

[15] G. Potje, F. Cadar, A. Araujo, R. Martins, and E. R. Nascimento, "XFeat: Accelerated features for lightweight image matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2682–2691.

[16] S. Zhang et al., "Multilevel attention Siamese network for keypoint detection in optical and SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5404617.

[17] Y. Xiang, F. Wang, and H. You, "OS-SIFT: A robust SIFT-like algorithm for high-resolution optical-to-SAR image registration in suburban areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3078–3090, Jun. 2018.

[18] J. Chen, J. Tian, N. Lee, J. Zheng, R. T. Smith, and A. F. Laine, "A partial intensity invariant feature descriptor for multimodal retinal image registration," *IEEE Trans. Biomed. Eng.*, vol. 57, no. 7, pp. 1707–1718, Jul. 2010.

[19] W. Ma et al., "Remote sensing image registration with modified SIFT and enhanced feature matching," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 1, pp. 3–7, Jan. 2017.

[20] Q. Yu, D. Ni, Y. Jiang, Y. Yan, J. An, and T. Sun, "Universal SAR and optical image registration via a novel SIFT framework based on nonlinear diffusion and a polar spatial-frequency descriptor," *ISPRS J. Photogramm. Remote Sens.*, vol. 171, pp. 1–17, Jan. 2021.

[21] Y. Ye and L. Shen, "HOPC: A novel similarity metric based on geometric structural properties for multi-modal remote sensing image matching," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 3, pp. 9–16, Jun. 2016.

[22] J. Li, Q. Hu, and M. Ai, "RIFT: Multi-modal image matching based on radiation-variation insensitive feature transform," *IEEE Trans. Image Process.*, vol. 29, pp. 3296–3310, 2020.

[23] J. Li, W. Xu, P. Shi, Y. Zhang, and Q. Hu, "LNIFT: Locally normalized image for rotation invariant multimodal feature matching," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5621314.

[24] Y. Yao, Y. Zhang, Y. Wan, X. Liu, X. Yan, and J. Li, "Multi-modal remote sensing image matching considering co-occurrence filter," *IEEE Trans. Image Process.*, vol. 31, pp. 2584–2597, 2022.

[25] C. Gao, W. Li, R. Tao, and Q. Du, "MS-HLMO: Multiscale histogram of local main orientation for remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626714.

[26] J. Li, Q. Hu, and Y. Zhang, "Multimodal image matching: A scale-invariant algorithm and an open dataset," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 77–88, Oct. 2023.

[27] H. Zhang et al., "Registration of multimodal remote sensing image based on deep fully convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3028–3042, Aug. 2019.

[28] M. Gazzea, O. Sommervold, and R. Arghandeh, "MARU-Net: Multiscale attention gated residual U-Net with contrastive loss for SAR-optical image matching," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4891–4899, 2023.

[29] S. Cui, A. Ma, Y. Wan, Y. Zhong, B. Luo, and M. Xu, "Cross-modality image matching network with modality-invariant feature representation for airborne-ground thermal infrared and visible datasets," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5606414.

[30] Y. Deng and J. Ma, "ReDFeat: Recoupling detection and description for multimodal feature learning," *IEEE Trans. Image Process.*, vol. 32, pp. 591–602, 2023.

[31] S. Cui, A. Ma, L. Zhang, M. Xu, and Y. Zhong, "MAP-Net: SAR and optical image matching via image-based convolutional network with attention mechanism and spatial pyramid aggregated pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 1000513.

[32] D. Quan et al., "Self-distillation feature learning network for optical and SAR image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4706718.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.

[33] D. Quan et al., "Efficient and robust: A cross-modal registration deep wavelet learning method for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4739–4754, 2023.
[34] Y. Zhang, C. Lan, H. Zhang, G. Ma, and H. Li, "Multimodal remote sensing image matching via learning features and attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603620.
[35] J. Li, X. Chen, X. Wang, G. Bi, T. Nie, and L. Huang, "AttFeat: Attention-based features for infrared and visible remote sensing image matching," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.
[36] Q. Xu, R. Zhang, Y. Zhang, Y. Wang, and Q. Tian, "A Fourier-based framework for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14378–14387.
[37] J. Wang, R. Du, D. Chang, K. Liang, and Z. Ma, "Domain generalization via frequency-domain-based feature disentanglement and interaction," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 4821–4829.
[38] S. Lin et al., "Deep frequency filtering for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11797–11807.
[39] B. Zitová and J. Flusser, "Image registration methods: A survey," *Image Vis. Comput.*, vol. 21, no. 11, pp. 977–1000, Oct. 2003.
[40] X. Tong et al., "Image registration with Fourier-based image correlation: A comprehensive review of developments and applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 10, pp. 4062–4081, Oct. 2019.
[41] Y. Zhang et al., "Histogram of the orientation of the weighted phase descriptor for multi-modal remote sensing image matching," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 1–15, Feb. 2023.
[42] G. Wan et al., "Multimodal remote sensing image matching based on weighted structure saliency feature," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4700816.
[43] D. Quan et al., "F3Net: Adaptive frequency feature filtering network for multimodal remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5222813.
[44] M. Li et al., "Cross-domain and cross-modal knowledge distillation in domain adaptation for 3D semantic segmentation," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 3829–3837.
[45] C. Hu, C. Zhang, and G. H. Lee, "Unsupervised feature representation learning for domain-generalized cross-domain image retrieval," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 11016–11025.
[46] Y. Ye, L. Bruzzone, J. Shan, F. Bovolo, and Q. Zhu, "Fast and robust matching for multimodal remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9059–9070, Nov. 2019.
[47] J. Fan, Y. Wu, M. Li, W. Liang, and Y. Cao, "SAR and optical image registration using nonlinear diffusion and phase congruency structural descriptor," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5368–5379, Sep. 2018.
[48] Z. Fan, L. Zhang, Y. Liu, Q. Wang, and S. Zlatanova, "Exploiting high geopositioning accuracy of SAR data to obtain accurate geometric orientation of optical satellite images," *Remote Sens.*, vol. 13, no. 17, p. 3535, Sep. 2021.
[49] L. H. Hughes, M. Schmitt, and X. X. Zhu, "Mining hard negative samples for SAR-optical image matching using generative adversarial networks," *Remote Sens.*, vol. 10, no. 10, p. 1552, Sep. 2018.
[50] Y. Ye, T. Tang, B. Zhu, C. Yang, B. Li, and S. Hao, "A multiscale framework with unsupervised learning for remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622215.
[51] Y. Ye, C. Yang, G. Gong, P. Yang, D. Quan, and J. Li, "Robust optical and SAR image matching using attention-enhanced structural features," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610212.
[52] S. Cui, M. Xu, A. Ma, and Y. Zhong, "Modality-free feature detector and descriptor for multimodal remote sensing image registration," *Remote Sens.*, vol. 12, no. 18, p. 2937, Sep. 2020.
[53] L. H. Hughes, D. Marcos, S. Lobry, D. Tuia, and M. Schmitt, "A deep learning framework for matching of SAR and optical imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 166–179, Nov. 2020.
[54] Y. Li et al., "A robust multisource remote sensing image matching method utilizing attention and feature enhancement against noise interference," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5651921.
[55] Y. Deng and J. Ma, "SDGMNet: Statistic-based dynamic gradient modulation for local descriptor learning," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 1510–1518.
[56] Y. Xiang, R. Tao, F. Wang, H. You, and B. Han, "Automatic registration of optical and SAR images via improved phase congruency model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5847–5861, 2020.
[57] P. Wu et al., "MapGlue: Multimodal remote sensing image matching," 2025, *arXiv:2503.16185*.
[58] L. Moisan and B. Stival, "A probabilistic criterion to detect rigid point matches between two images and estimate the fundamental matrix," *Int. J. Comput. Vis.*, vol. 57, no. 3, pp. 201–218, May 2004.
[59] X. Jia, C. Zhu, M. Li, W. Tang, and W. Zhou, "LLVIP: A visible-infrared paired dataset for low-light vision," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 3489–3497.
[60] J. Cho, D. Min, Y. Kim, and K. Sohn, "DIML/CVL RGB-D dataset: 2M RGB-D images of natural indoor and outdoor scenes," 2021, *arXiv:2110.11590*.



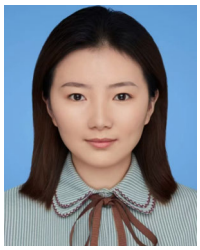
**Peifu Lai** received the B.S. degree from the School of Automation, Southeast University, Nanjing, China, in 2023, where he is currently pursuing the M.S. degree.

His research interests include multimodal feature learning and its applications.



**Yaping Yan** received the B.S. degree in communication engineering from Zhejiang University, Hangzhou, China, in 2010, the M.S. degree in circuits and systems from Lanzhou University, Lanzhou, China, in 2016, and the Ph.D. degree in information science from Hokkaido University, Sapporo, Japan, in 2020.

She is currently with the School of Computer Science and Engineering, Southeast University, Nanjing, China. Her researches focus on pattern recognition and computer vision.



**Na Zhang** received the M.S. degree from Beijing Institute of Technology, Beijing, China, in 2022.

Her research focuses on visual matching and navigation.



**Songlin Du** (Senior Member, IEEE) received the Ph.D. degree in physics from Lanzhou University, Lanzhou, China, and Ph.D. degree in Engineering from Waseda University, Tokyo, Japan, both in 2019.

He is currently an Associate Professor with the School of Automation, Southeast University, Nanjing, China. He is the co-author of the book *Human Pose Analysis* (Singapore: Springer Nature, 2024). His research interests include computer vision and its applications.

Dr. Du was a recipient of the Best Paper Award from the IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS) in 2017, the Best Presentation Award from the International Conference on Intelligent Systems and Image Processing (ICISIP) in 2018, the Excellent Paper Award from the International Collaboration Symposium on Information, Production and System (ISIPS) in 2018, and the Best Presentation Award from the Computer Vision Session of the International Conference on Artificial Intelligence and Virtual Reality (AIVR) in 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:46:27 UTC from IEEE Xplore. Restrictions apply.